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Final report

University *of Ljubljana*
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RAG assistant for Slovenian employment law

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Abstract

We built a retrieval-augmented generation assistant for Slovenian employment-law questions. The final system intentionally uses one main approach, RAG, because the target machine is CPU-limited and legal correctness depends more on current official sources than on model memorization. The assistant retrieves curated legal passages, applies a strict Slovenian system prompt, cites the retrieved source, and refuses unsupported or out-of-domain questions. We evaluated retrieval separately from generation, monitored official source availability, and compared a curated primary-law corpus with a larger COLESLAW-derived extraction. The best current setup is strict RAG over verified primary-law chunks: retrieval reaches 0.938 answerable hit rate with 0.000 false-evidence rate on unanswerable questions.

Keywords

Slovenian employment law, legal question answering, retrieval-augmented generation, RAG, GaMS

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Motivation

General-purpose LLMs are risky for legal question answering because they may answer fluently while inventing legal rules, citations, or article numbers. Our domain is therefore restricted to Slovenian employment law: employment contracts, termination, leave, wages, working time, sick leave, collective employment issues, and closely related statutory guidance. The assistant is informational, not a substitute for a lawyer, and must refuse tax, inheritance, company-formation, foreign-law, or other non-employment-law questions.

Related Work

Existing Slovenian legal assistants such as Pravko and MojMaliPravnik show demand for accessible legal explanations, but they do not give us a reproducible course pipeline or a controlled official-source policy [1, 2]. For knowledge injection into LLMs we considered prompt engineering, RAG, fine-tuning, and tool use. RAG is well suited to knowledge-intensive tasks because claims can be grounded in retrieved documents [3]. LoRA-style fine-tuning is attractive for style or task adaptation [4], while tool-using models can call external systems [5]; however, both add implementation and hardware cost. Following the survey perspective that domain knowledge can be injected at several stages [6], we chose the simplest auditable path: source curation, retrieval, prompt guardrails, and evaluation before any fine-tuning.

Corpus and Sources

We used COLESLAW 1.0 as the broad legal starting point [7], but the final RAG corpus is source-prioritized rather than corpus-size-maximized. COLESLAW is valuable for case-law demonstrations, yet our 500-chunk employment-law extraction was dominated by *SodnaPraksa* and old collective-agreement material. For statutory question answering this caused weaker retrieval than the smaller curated primary-law set.

The final source policy is strict. PISRS statutes, especially ZDR-1, are canonical primary law [8]. GOV.SI, MDDSZ, IRSD, eUprava, SPOT, ESS, and OPSI are official interpretation or operational sources [9, 10]. *sodnapraksa.si* and selected COLESLAW case-law chunks are tertiary support only and may not override current statutes. The source monitor currently checks 12 PISRS sources, 17 government or official interpretation sources, and one official case-law index; the latest run found all reachable.

System

The implemented pipeline is RAG-first. Documents are split into compact chunks with metadata such as law name, article, title, and source. The current retriever is BM25 with a simple lexical fallback, which keeps the project runnable on CPU and without a vector database. At answer time the top retrieved chunks are formatted as context and passed to the generator with the selected prompt `strict_legal_rag_sl_v2`. The prompt requires Slovenian-only answers, checks Slovenian employment-law scope, prioritizes PISRS over official interpretations and case law, and uses a fixed refusal sentence when the context is insufficient.

The preferred model for the next live deployment is GaMS-1B-Chat, because it is Slovenian-focused and CPU-friendlier than larger GaMS variants [11]. Until GaMS is served through Open WebUI or a compatible local endpoint, the repository keeps deterministic offline tests and Ollama/Open WebUI fallbacks. Fine-tuning data preparation exists only as an exploratory artifact; it is not used for the final claim.

Evaluation

We evaluate retrieval and answer quality separately. Retrieval metrics are answerable hit rate, false-evidence rate for unanswerable questions, and average context length. Answer metrics are correctness, grounding, completeness, clarity, hallucination, and refusal accuracy. The judge has an offline fallback so that the repository remains reproducible when no local model server is available.

Table 1. Final RAG optimization summary. Higher is better except false evidence, context length, and hallucination.

Configuration	Hit	False ev.	Ctx. words
Curated RAG	**0.938**	**0.000**	**191.8**
COLESLAW sample	0.500	0.250	557.2

Table 2. Offline smoke evaluation on the corrected curated corpus.

Variant	Correctness	Hallucination	Refusal
Baseline	0.70	4.20	0.00
Strict RAG	**3.20**	**0.85**	**1.00**

The main result is that corpus quality matters more than corpus size. The curated legal chunks retrieve the expected law for nearly all answerable questions and avoid false support for out-of-scope questions. The larger COLESLAW extraction improves breadth, but without source-priority filtering it retrieves too much case law for statutory questions. The strict prompt improves refusal and hallucination behavior on the curated corpus. On the COLESLAW optimization sample, strict RAG still improved correctness from 0.70 to 1.85 and reduced hallucination from 4.20 to 1.80, but the lower score confirms that prompting cannot compensate for weak retrieval context.

Limitations and Future Work

The project does not yet implement a production standalone app; the app design is documented separately and recommends a PWA before Tauri or Flutter. The next highest-impact RAG work is article-level PISRS ingestion, source-priority reranking, citation validation, and expansion of the evaluation set with more current-law and out-of-scope examples. Dense embeddings and reranking are promising, but should be added only after the official corpus is clean and monitored. Fine-tuning remains a future option if RAG and prompting stop improving grounded correctness.

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